

2 Methodology

2.1 Causal framework

We consider data from two sources: a randomised controlled trial (RCT) and a real-world data study (RWD). The data-source indicator $S \in \{0, 1\}$ takes the value $S = 1$ for RCT observations and $S = 0$ for RWD observations. The RCT contributes n_1 independent observations and the RWD contributes n_0 independent observations. Each observation has a vector of pre-treatment covariates $X \in \mathbb{R}^p$ and receives a binary treatment $A \in \{0, 1\}$. The outcome of interest is the nonnegative survival time T , subject to censoring. We allow three forms of censoring: exact observation, right-censoring, and interval-censoring. For each subject we observe a pair (L, R) with $0 \leq L \leq R \leq \infty$ satisfying $T \in [L, R]$, together with a censoring-type indicator $\delta \in \{0, 1, 2\}$. We use the convention that $R = \infty$ stands for $T \in [L, \infty)$. These correspond to exact observation ($\delta = 1$: $L = R = T$), right-censoring ($\delta = 0$: $L = C$ and $R = \infty$, for some censoring time C), and interval-censoring ($\delta = 2$: $0 \leq L < R < \infty$). The standard right-censored setup is recovered when $\delta \in \{0, 1\}$ for every subject. We write $\mathcal{C}$ for the inspection and censoring process underlying (L, R, δ) , and refer to [Sun \(2006\)](#) for background. We adopt the potential-outcomes framework ([Rubin, 1974](#)): $T(a)$, $\mathcal{C}(a)$, and the corresponding observations $(L(a), R(a), \delta(a))$ are defined analogously under treatment $A = a$.

We estimate the conditional average treatment effect (CATE):

$$\tau(x) = \mathbb{E}[\log T(1) - \log T(0) \mid X = x]. \quad (1)$$

The CATE captures the heterogeneity of the treatment effect across covariate values. The acceleration factor (AF) is the corresponding estimand on the multiplicative time scale:

$$\text{AF}(x) = \exp\{\tau(x)\}. \quad (2)$$

The acceleration factor admits a direct interpretation on the survival-time scale: at every quantile of the survival distribution, treatment rescales the survival time by a constant multiplicative factor ([Brathovde et al., 2024](#)). This time-scaling is more intuitive in a clinical setting than the relative change in event rate conveyed by a hazard ratio ([Swindell, 2009](#)). The acceleration factor also enjoys causal properties that the hazard ratio lacks. [Brathovde et al. \(2024\)](#) show that the observed acceleration factor identifies its causal counterpart under exchangeability and consistency. This identification continues to hold under unmeasured frailty and treatment-effect heterogeneity. The hazard ratio loses its causal interpretation in those same settings, because conditioning on survival induces a built-in selection bias ([Hernán, 2010](#)). The acceleration factor is collapsible: the marginal and conditional acceleration factors coincide when the treatment effect is homogeneous, regardless of the distribution of unmeasured baseline risk ([Aalen et al., 2015](#); [Crowther et al., 2023](#)).

We impose conditions on the causal structure of the problem and on the censoring mechanism.

Assumption 1 (Consistency). $T = AT(1) + (1 - A)T(0)$.

Assumption 2 (Unconfoundedness of the RCT). $T(a) \perp\!\!\!\perp A \mid X, S = 1$ for each $a \in \{0, 1\}$.

Assumption 3 (Positivity). $0 < \mathbb{P}(A = 1 \mid X = x, S = s) < 1$ for each $s \in \{0, 1\}$ and all x in the support of $X \mid S = s$.

Assumption 4 (Cross-source exchangeability). $T(a) \mid X, S = 1 \stackrel{d}{=} T(a) \mid X, S = 0$, for each $a \in \{0, 1\}$.

Assumption 5 (Conditionally non-informative censoring). $T(a) \perp\!\!\!\perp \mathcal{C}(a) \mid X, A, S$ for each $a \in \{0, 1\}$.

Remark 1 (Possible weakening to CATE-only transport). Assumption 4 may be stronger than needed. It could be replaced by the weaker condition that only the conditional log-treatment effect transports across sources:

$$\mathbb{E}[\log T(1) - \log T(0) \mid X, S = 1] = \mathbb{E}[\log T(1) - \log T(0) \mid X, S = 0]. \quad (3)$$

This alleviation may suffice in our setting because the causal estimand is the CATE on the log-time scale. *To be verified.*

Our identification strategy relaxes unconfoundedness in the RWD and retains transportability of the CATE across sources. Assumption 2 holds by design in a randomised trial. Treatment in the RWD may depend on confounders not contained in X . The resulting bias is absorbed by the confounding function $c(x)$ introduced in the next section. We assume the CATE to be time-invariant by not explicitly letting it depend on time. Other approaches that combine evidence from multiple sources take the opposite route: they allow source-specific CATEs but require unconfoundedness within each source (Stuart et al., 2011; Dahabreh et al., 2019; Cheng and Cai, 2021; Shyr et al., 2025). Such heterogeneity in the CATE arises from varying distributions of unobserved effect modifiers (Degtiar and Rose, 2023), for example physical activity in breast-cancer treatment (Cannioto et al., 2021). The two relaxations cannot coexist in our setting. Without unconfoundedness of the RWD, the CATE and $c(x)$ are not separately identifiable unless the CATE transports across sources. We therefore retain Assumption 4 as the explicit cost of absorbing RWD confounding into $c(x)$, and acknowledge that CATE heterogeneity across sources falls outside the scope of the current framework.

2.2 Accelerated failure time decomposition

The confounding function captures the bias in the RWD that the causal effect τ alone cannot explain. Unconfoundedness fails in the RWD by design. The treated-versus-control contrast observed in the RWD therefore differs from the CATE in general. We define the *confounding function* $c(x)$ as:

$$c(x) = \mathbb{E}[\log T \mid X = x, A = 1, S = 0] - \mathbb{E}[\log T \mid X = x, A = 0, S = 0] - \tau(x). \quad (4)$$

The confounding function represents the difference between the RWD treated-versus-control contrast and the true causal effect. Under Assumptions 1–4, the conditional log survival time admits an additive decomposition into a baseline, the causal effect, and the confounding bias.

Proposition 1 (Accelerated failure time decomposition). *Under Assumptions 1–4, the conditional log survival time satisfies:*

$$\mathbb{E}[\log T \mid A, X, S] = m_0(X, S) + \tau(X) A + (1 - S) A c(X), \quad (5)$$

where $m_0(X, S) := \mathbb{E}[\log T \mid A = 0, X, S]$ is the baseline log survival time.

The decomposition separates three pieces. The baseline $m_0(X, S)$ is the control-arm log survival time in each source. The CATE $\tau(X)$ enters additively whenever an observation is treated, regardless of source. The confounding function $c(X)$ contributes only in the treated arm of the RWD, reflecting that the RWD treated-versus-control contrast carries bias on top of the causal effect.

Proposition 2 (Identification). *Under Assumptions 1–4:*

(i) *The CATE is identified from the RCT alone:*

$$\tau(x) = \mathbb{E}[\log T \mid X = x, A = 1, S = 1] - \mathbb{E}[\log T \mid X = x, A = 0, S = 1]. \quad (6)$$

(ii) *Given τ , the confounding function is identified from the RWD.*

(iii) *Neither τ nor c is identified from the RWD alone; the RWD identifies only the composite $\tau(x) + c(x)$.*

Proofs of Propositions 1 and 2 are in Supplementary Materials 2.7. The same identification result holds for the acceleration factor $\text{AF}(x)$ by continuity of the exponential.

We illustrate the confounding function through a structural causal model with an unmeasured confounder U as a special case in Supplementary Materials 2.8. This reading gives $c(x)$ a mechanistic interpretation. It decomposes $c(x)$ into a prognostic imbalance between the RWD treatment arms and effect modification by U . The decomposition motivates the prior placed on c .

We model the log survival time with the accelerated failure time (AFT) specification implied by the decomposition:

$$\log T_i = m_0(X_i, S_i) + A_i \tau(X_i) + (1 - S_i) A_i c(X_i) + \varepsilon_i. \quad (7)$$

We model the errors ε_i as independent with mean zero and a nonparametric distribution. The regression functions m_0 , τ and c retain the causal meanings established above. We assign each a separate Bayesian tree ensemble prior.

2.3 Nonparametric error distribution

We model the errors ε_i in (7) with a hierarchical Dirichlet process mixture of normals (HDPM). The prior is source-specific: $\varepsilon_i \mid S_i = s \sim F_s$. The model lets F_0 and F_1 differ flexibly while sharing information about which residual types are present across sources. Existing data-fusion methods typically assume $F_0 = F_1$, for example via a single Gaussian likelihood. This assumption is unsatisfactory in our setting. Randomised trials and real-world data sources typically differ in measurement protocol, follow-up quality, and the prevalence of extreme outcomes. These differences plausibly enter the residual distribution rather than the structural mean. The two error distributions are unlikely to be entirely unrelated. A model that treats them as fully independent forfeits useful pooling, particularly when one source is small.

Conditional on the source indicator $S_i = s$, we model the residual as a location mixture of Gaussians with source-specific scale $\sigma_s > 0$ and mixing measure G_s :

$$\varepsilon_i \mid S_i = s, G_s, \sigma_s \sim \int \frac{1}{\sigma_s} \phi\left(\frac{w - \theta}{\sigma_s}\right) dG_s(\theta). \quad (8)$$

Gaussian-mixture priors of this form have been used for AFT residuals in single-source BART models (Henderson et al., 2020). The mixing measures share latent structure through the hierarchical Dirichlet process of Teh et al. (2006), with a top-level random measure G^* :

$$G^* \sim \text{DP}(\gamma, H), \quad G_s^* \mid G^*, M_s \sim \text{DP}(M_s, G^*), \quad s \in \{0, 1\}, \quad (9)$$

with base distribution $H = N(0, \sigma_\theta^2)$. By Sethuraman’s stick-breaking representation (Sethuraman, 1994), the top-level measure is discrete:

$$G^* = \sum_{k=1}^{\infty} \beta_k \delta_{\theta_k^*}, \quad \theta_k^* \stackrel{\text{iid}}{\sim} H, \quad \beta_k = u_k \prod_{l < k} (1 - u_l), \quad u_k \stackrel{\text{iid}}{\sim} \text{Beta}(1, \gamma). \quad (10)$$

Each source-specific measure G_s^* is supported on the same atoms $\{\theta_k^*\}$ as G^* , with its own weight vector $\boldsymbol{\pi}_s = (\pi_{sk})_{k \geq 1}$. The shape of each F_s is therefore parameterised by a shared set of residual types and a source-specific weighting over them.

For m_0 , τ , and c to retain their interpretation as the conditional-mean components defined in (5), each F_s must have mean zero. We follow Yang et al. (2010) and obtain centred mixing measures by subtracting the source-specific mean of the unconstrained measure:

$$\mu_s = \int \theta dG_s^*(\theta) = \sum_{k=1}^{\infty} \pi_{sk} \theta_k^*, \quad G_s = \sum_{k=1}^{\infty} \pi_{sk} \delta_{\theta_k^* - \mu_s}. \quad (11)$$

By construction $\mathbb{E}[\varepsilon_i \mid S_i = s, G_s, \sigma_s] = 0$. The unconstrained atoms θ_k^* remain shared across sources, while the centred atoms $\theta_k^* - \mu_s$ are source-specific through the shifts μ_0 and μ_1 . We place a shared scaled-inverse- χ^2 prior $\sigma_s^2 \sim \nu\lambda/\chi_\nu^2$ on the per-source residual scales, with degrees of freedom ν and scale λ . We set $\nu = 3$ following the BART default and calibrate λ to the empirical residual variance following Chipman et al. (2010).

2.4 Overview of BART

Bayesian additive regression trees (Chipman et al., 2010) provide a flexible nonparametric prior over an unknown function $f : \mathbb{R}^p \rightarrow \mathbb{R}$. The prior represents f as a sum of regression trees:

$$f(X) = \sum_{j=1}^m g(X; \mathcal{T}_j, \mathcal{H}_j), \quad (12)$$

where each $\mathcal{T}_j$ is a binary tree with b_j terminal nodes. Each interior splitting rule has the form $\{x_\ell \leq c\}$ for some covariate index ℓ and splitting value c . We write $\mathcal{H}_j = \{h_{j,1}, \dots, h_{j,b_j}\}$ for the step heights at those terminal nodes. The function $g(X; \mathcal{T}_j, \mathcal{H}_j)$ routes X through the splits of $\mathcal{T}_j$ to a terminal node a and returns the step height $h_{j,a}$ at that node.

The prior on $(\mathcal{T}_1, \mathcal{H}_1), \dots, (\mathcal{T}_m, \mathcal{H}_m)$ is independent across trees. It decomposes into a tree-structure prior and a step-height prior given the structure. The structure of each $\mathcal{T}_j$ follows the recursive splitting process of Chipman et al. (1998). A node at depth d is non-terminal with probability $\alpha(1 + d)^{-\beta}$. This generative process is a heterogeneous Galton–Watson branching process, in which the per-generation offspring distribution depends on the depth d (Ročková and Saha, 2019). The hyperparameters α and β regularise tree depth. Conditional on the tree, the splitting variable x_ℓ at each interior node is drawn uniformly from the available covariates. The splitting value c is drawn uniformly from its observed values. The step heights receive independent conjugate normal priors $h_{j,a} \mid \mathcal{T}_j \sim \mathcal{N}(0, \sigma_h^2)$, parametrised as $\sigma_h = k/(2\sqrt{m})$ with $k > 0$ the step-height scale. Chipman et al. (2010) centre and rescale the response, then choose k so that the implied prior on f assigns about 95% of its mass to the observed response range. We denote the BART prior with these hyperparameters by $\text{BART}(m, k, \alpha, \beta)$. The four arguments are the number of trees m , the step-height scale k , and the tree-structure hyperparameters α and β .

2.5 Prior specification

We place independent Bayesian tree priors on the components of the decomposition (5). We tune the regularisation of each component through the tree-structure prior and the ensemble size to match the substantive role of each component.

We allow for heterogeneity in the baseline effect between sources. The prognostic function m_0 is split into a shared component and a source-specific deviation so that the prior centres on cross-source borrowing. We write:

$$m_0(X, S) = m_0^{\text{sh}}(X) + (1 - S) d(X), \quad (13)$$

where m_0^{sh} is shared across sources and d is active only in the RWD. The decomposition writes the difference between the two source-specific prognostic functions as $d(X)$. The prior on d has mean zero, so the prior centres on equality of the two prognostic functions. An alternative would place a single BART on $m_0(X, S)$ with S as an ordinary covariate. Such a model does not encode this prior centring. The prior on d then provides a direct dial for the strength of borrowing. We use the tree-structure hyperparameters $(\alpha_{\text{sh}}, \beta_{\text{sh}}) = (0.95, 2)$ and $m_{\text{sh}} = 200$ trees for m_0^{sh} . The large ensemble lets m_0^{sh} accommodate complex disease-intrinsic structure. We use the same hyperparameters $(\alpha_d, \beta_d) = (0.95, 2)$ for d but only $m_d = 50$ trees. The smaller ensemble suffices because d only needs to absorb residual differences in protocol and patient population. This construction coincides with a heterogeneous meta-analytic-predictive prior on the prognostic function (Supplementary Materials 2.9).

We place a depth-penalised BART prior on the treatment-effect function τ to discourage higher-order interactions in X . We set the depth parameter $\beta_\tau = 3$ and $\alpha_\tau = 0.95$. The prior mass concentrates on trees of depth one or two. The number of trees is reduced relative to the prognostic baseline to $m_\tau = 100$ trees. This reflects the expectation that the treatment-effect surface is simpler than the prognostic baseline.

We place a more strongly regularised BART prior on the confounding function c . We set the base splitting probability $\alpha_c = 0.25$ and the depth parameter $\beta_c = 3$. We set the number of trees to $m_c = 50$. This discourages splits at the root and deep interactions when splits do occur.

Remark 2 (Multiplicative shrinkage on c). An alternative representation is $c(x) = \gamma g_c(x)$, with γ a scalar shrinkage parameter and $g_c \sim \text{BART}$. A heavy-tailed prior such as $\gamma \sim \mathcal{C}^+(0, 1)$ concentrates the posterior at small magnitudes while permitting occasional large values. A normal prior $\gamma \sim \mathcal{N}(0, \sigma_\gamma^2)$ is a softer alternative. Either choice imposes stronger regularisation on $c(x)$ than the sole BART prior, and the posterior of γ admits a reading as a level of RWD confounding.

We place a shared scaled-inverse- χ^2 prior $\sigma_s^2 \sim \nu\lambda/\chi_\nu^2$ on the per-source residual scales with ν degrees of freedom and scale λ . We set $\nu = 3$ following the BART default and calibrate λ to the empirical residual variance following Chipman et al. (2010). We standardise the log survival time before fitting by the centring and scaling constants of a preliminary log-normal AFT fit. This rescales the response to a unit-variance scale. Each forest carries its own prior step-height scale k_f with $f \in \{m_0^{\text{sh}}, d, \tau, c\}$. The step-height scales calibrate the prior magnitude of each ensemble to match the scale of the standardised response. We recommend $k_f = 1$ for all four forests as a default and allow further tuning via cross-validation.

We summarise the full hierarchical model:

$$\begin{aligned}
\log T_i &= m_0^{\text{sh}}(X_i) + (1 - S_i) d(X_i) + A_i \tau(X_i) + (1 - S_i) A_i c(X_i) + \varepsilon_i, \\
m_0^{\text{sh}} &\sim \text{BART}(200, k_{\text{sh}}, 0.95, 2), \\
d &\sim \text{BART}(50, k_d, 0.95, 2), \\
\tau &\sim \text{BART}(100, k_\tau, 0.95, 3), \\
c &\sim \text{BART}(50, k_c, 0.25, 3), \\
\varepsilon_i \mid S_i = s, G_s, \sigma_s &\sim \int \sigma_s^{-1} \phi\left(\frac{w - \theta}{\sigma_s}\right) dG_s(\theta), \\
G_s &= \sum_k \pi_{sk} \delta_{\theta_k^* - \mu_s}, \quad s \in \{0, 1\}, \\
G_s^* \mid G^*, M_s &\sim \text{DP}(M_s, G^*), \quad s \in \{0, 1\}, \\
G^* &\sim \text{DP}(\gamma, H), \quad H = N(0, \sigma_\theta^2), \\
\sigma_s^2 &\sim \nu \lambda / \chi_\nu^2, \quad s \in \{0, 1\}.
\end{aligned}$$

2.6 Posterior inference*

This section is largely unfinished. The paragraphs on average treatment effects and posterior computation are ready. The following items are ideas for material that could go in this section.

- **Treatment-effect estimands.** Posterior summaries of the CATE $\tau(x)$ and the acceleration factor $\text{AF}(x) = \exp\{\tau(x)\}$ at fixed x : posterior mean and credible bands. Investigate the impact of covariates on the outcome, e.g. via linear projections or TE-VIM's.
- **Heterogeneity.** Posterior measures of heterogeneity of the treatment effect.
- **Borrowing.** Posterior measures of borrowing: behaviour of the deviation forest d relative to the shared baseline m_0^{sh} , magnitude of the borrowing scale σ_d , posterior mass on $c \equiv 0$ et cetera.
- **Residual fit.** Was the heavy nonparametric model for the error distribution justified, or would a simpler model suffice? Posterior number of active mixture clusters and cross-source comparison of F_0 and F_1 for example.

We compute average treatment effects by marginalising the conditional effect $\tau(\cdot)$ against a target covariate distribution. We take the target to be the union of the two source studies, with covariate distribution $F = \pi_0 F_0 + \pi_1 F_1$. Here F_s is the covariate distribution within source $s \in \{0, 1\}$ and $\pi = (\pi_0, \pi_1)$ lies on the unit simplex. The posterior of the marginal effect carries uncertainty from three sources: the posterior over τ , the within-source covariate distributions F_s , and the mixing fractions π . We propagate all three jointly by a hierarchical Bayesian bootstrap. The choice of π encodes the analyst's target population. Natural deterministic choices include the empirical fractions $\pi_s = n_s/n$ and equal weighting $\pi = (1/2, 1/2)$. The source-specific limits $\pi = (1, 0)$ and $\pi = (0, 1)$ target the population of the RWD and RCT respectively. We may instead place a Dirichlet prior $\pi \sim \text{Dirichlet}(\alpha)$ on the mixing fractions, with concentration parameter $\alpha = (\alpha_0, \alpha_1)$. This propagates uncertainty about the target composition. The choice $\alpha = (n_0, n_1)$ centres the prior on the empirical fractions and approximately recovers a pooled Bayesian bootstrap. We recommend this as the default. The choice $\alpha = (1, 1)$ gives a flat prior on the simplex. We draw mixing fractions $\pi^{(b)} \sim \text{Dirichlet}(\alpha)$ for each posterior draw $\tau^{(b)}(\cdot)$ of the conditional effect. We draw within-source weights $w^{(s,b)} \sim \text{Dirichlet}(\mathbf{1}_{n_s})$ over the

observations of source s independently for $s \in \{0, 1\}$. The corresponding draw of the average treatment effect is:

$$\bar{\tau}^{(b)} = \sum_{s \in \{0, 1\}} \pi_s^{(b)} \sum_{i: S_i = s} w_i^{(s, b)} \tau^{(b)}(X_i). \quad (14)$$

The within-source weights are the Bayesian bootstrap of [Rubin \(1981\)](#) applied to each F_s .

We sample from the posterior via a blocked Gibbs sampler built on an extended Bayesian backfitting approach ([Hastie and Tibshirani, 2000](#)). We update the BART forests m_0^{sh} , d , τ , and c in turn against the partial residual. The partial residual subtracts the other forests and the current draw of the residual mixture from the augmented outcome. Each update reduces to a standard BART update on a Gaussian working response ([Chipman et al., 2010](#)). We augment the censored observations ([Tanner and Wong, 1987](#)). The HDPM error block is updated by a separate Gibbs step over its cluster assignments, atoms, mixture weights, and concentration parameters. We give the full sampler in the Supplementary Materials.

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2.7 Supplementary Materials: Proofs

Proof of Proposition 1. We evaluate $\mathbb{E}[\log T \mid A, X, S]$ for each value of (A, S) .

Case $A = 0$. By the definition of m_0 , the right-hand side of (5) reduces to $m_0(X, S)$, which matches the left-hand side.

Case $A = 1, S = 1$ (RCT). Consistency (Assumption 1) gives $\log T = \log T(1)$ when $A = 1$, and Assumption 2 gives $T(1) \perp\!\!\!\perp A \mid X, S = 1$, so:

$$\mathbb{E}[\log T \mid A = 1, X, S = 1] = \mathbb{E}[\log T(1) \mid X, S = 1]. \quad (15)$$

We split this conditional expectation as:

$$\mathbb{E}[\log T(1) \mid X, S = 1] = \mathbb{E}[\log T(0) \mid X, S = 1] + \mathbb{E}[\log T(1) - \log T(0) \mid X, S = 1]. \quad (16)$$

The first term equals $m_0(X, 1)$ by Assumption 2. The second term equals $\tau(X)$ by Assumption 4.

Case $A = 1, S = 0$ (RWD). The definition of c in (4) rearranges to:

$$\mathbb{E}[\log T \mid A = 1, X, S = 0] = m_0(X, 0) + \tau(X) + c(X), \quad (17)$$

which matches the right-hand side of (5). $\square$

Proof of Proposition 2. Part (i) follows by taking the difference of the $A = 1, S = 1$ and $A = 0, S = 1$ cases of (5). Part (ii) is immediate from rearranging (4). Part (iii) follows because (5) restricted to $S = 0$ identifies only $m_0(X, 0)$ and the composite $\tau(X) + c(X)$. $\square$

2.8 Supplementary Materials: Structural interpretation of the confounding function

Context and goal. The main text treats the RWD as potentially confounded without specifying the source of the confounding. The confounding function $c(x)$ was defined in (4) as the residual bias in the RWD treated-versus-control contrast that the causal effect τ does not explain. This is an operational definition: it says how to compute c from the observed conditional means, but it says nothing about what real-world mechanisms generate the bias.

The aim of this section is to give $c(x)$ a causal-mechanistic interpretation. We achieve this by introducing a latent random vector $U \in \mathcal{U}$, called the *unmeasured confounder*, which stands in for any prognostic factor that affects survival but was not recorded in X . Examples include frailty, comorbidities, biomarkers, performance status, genotype, and behavioural risk. Under a structural model that lets U enter the baseline and the treatment effect, the confounding function decomposes into two interpretable terms: an imbalance in the baseline prognostic function between the RWD treatment arms, and a term capturing effect modification by U .

The decomposition has three uses. First, it makes precise what $c(x)$ measures in causal terms. Second, it identifies the conditions under which $c(x)$ vanishes. This in turn motivates the shrinkage prior placed on c in the modelling section. Third, it provides an entry point for sensitivity analyses that bound the magnitude of c in terms of properties of U .

Structural outcome model. We posit an outcome equation on the log-time scale:

$$\log T(a) = f_0(X, U) + a\tau^*(X, U) + \varepsilon, \quad \varepsilon \perp\!\!\!\perp (X, U, A, S). \quad (18)$$

The components have the following meanings. $X \in \mathcal{X}$ is the vector of observed pre-treatment covariates introduced in the main text. $U \in \mathcal{U}$ is the latent vector of unmeasured covariates described above; it is unknown to the analyst and never enters the likelihood used for inference. $f_0(X, U)$ is the baseline log-transformed survival under control, that is, the conditional mean of $\log T(0)$ given (X, U) . $\tau^*(X, U) = \mathbb{E}[\log T(1) - \log T(0) \mid X, U]$ is the treatment effect on the log scale conditional on the full covariate vector (X, U) . ε is an exogenous mean-zero error, independent of (X, U, A, S) ; its distribution is left unspecified and follows the same flexible nonparametric form used in the main text. The CATE on the log scale arises as the conditional mean of τ^* over U given X :

$$\tau(x) = \mathbb{E}[\tau^*(X, U) \mid X = x]. \quad (19)$$

The model imposes additive separability of the baseline and the treatment effect. It does not restrict the joint distribution of (X, U, A) in the RWD, so the RWD remains free to be confounded through U .

Structural form of $c(x)$.

Proposition 3 (Structural form of the confounding function). *Under the structural model (18) and Assumptions 1–4, the confounding function admits the decomposition:*

$$c(x) = \underbrace{\mathbb{E}[f_0(X, U) \mid X = x, A = 1, S = 0] - \mathbb{E}[f_0(X, U) \mid X = x, A = 0, S = 0]}_{\text{prognostic imbalance}} + \underbrace{\delta_\tau(x)}_{\text{effect modification by } U}, \quad (20)$$

where:

$$\delta_\tau(x) = \mathbb{E}[\tau^*(X, U) \mid X = x, A = 1, S = 0] - \tau(x). \quad (21)$$

Proof. Consistency (Assumption 1) gives $\log T = \log T(A)$ for every observation. We evaluate the two conditional expectations entering the definition (4) of the confounding function. In the RWD treated arm:

$$\mathbb{E}[\log T \mid X = x, A = 1, S = 0] = \mathbb{E}[f_0(X, U) + \tau^*(X, U) + \varepsilon \mid X = x, A = 1, S = 0], \quad (22)$$

and in the RWD control arm:

$$\mathbb{E}[\log T \mid X = x, A = 0, S = 0] = \mathbb{E}[f_0(X, U) + \varepsilon \mid X = x, A = 0, S = 0]. \quad (23)$$

The error term contributes zero in both expressions because ε has mean zero and is independent of (X, U, A, S) . The difference of the two expectations is

$$\begin{aligned} \mathbb{E}[\log T \mid X = x, A = 1, S = 0] - \mathbb{E}[\log T \mid X = x, A = 0, S = 0] \\ = \mathbb{E}[f_0(X, U) \mid X = x, A = 1, S = 0] - \mathbb{E}[f_0(X, U) \mid X = x, A = 0, S = 0] \\ + \mathbb{E}[\tau^*(X, U) \mid X = x, A = 1, S = 0]. \end{aligned} \quad (24)$$

We subtract $\tau(x)$ from both sides and apply the definition (4) of $c(x)$. The result is (20), with $\delta_\tau(x)$ as in (21). $\square$

Reduction under no unobserved effect modification. The structural form simplifies sharply when the treatment effect does not depend on U .

Corollary 1 (Reduction to prognostic imbalance). *If $\tau^*(X, U)$ depends on the covariates only through X , i.e. $\tau^*(X, U) = \tau(X)$, then $\delta_\tau(x) = 0$ and the confounding function reduces to:*

$$c(x) = \mathbb{E}[f_0(X, U) \mid X = x, A = 1, S = 0] - \mathbb{E}[f_0(X, U) \mid X = x, A = 0, S = 0]. \quad (25)$$

Proof. If $\tau^*(X, U) = \tau(X)$, then $\mathbb{E}[\tau^*(X, U) \mid X = x, A = 1, S = 0] = \tau(x)$, so $\delta_\tau(x) = 0$. Substitution into (20) yields (25). $\square$

Interpretation of the two components. The two pieces of (20) have distinct substantive meanings.

The *prognostic imbalance* term reflects that treated and control observations in the RWD differ in their distribution of U given X . Such an imbalance arises whenever treatment assignment in the RWD depends on unmeasured prognostic factors. For example, suppose U encodes frailty. At the same observed covariate value x , frailer patients are less likely to be assigned the treatment. Then U is on average lower in the RWD treated arm than in the RWD control arm at the same x , and the conditional expectations of $f_0(X, U)$ differ across arms. The sign and magnitude of the imbalance depend on the strength of the U - A association and on the dependence of f_0 on U .

The term $\delta_\tau(x)$ captures *effect modification by U* . It is non-zero only when τ^* itself depends on U and the distribution of U given X in the treated RWD arm differs from that in the population. For intuition, suppose treatment works better for patients with a certain unmeasured trait. If the RWD selectively treats patients with that trait, the RWD treated arm exhibits a stronger effect than the population CATE. This piece is identically zero under Corollary 1.

When does the confounding function vanish? By Proposition 3, $c(x) = 0$ at x whenever both the prognostic imbalance and $\delta_\tau(x)$ vanish. A simple sufficient condition is the unmeasured-confounder independence $U \perp\!\!\!\perp A \mid X, S = 0$: the unmeasured covariate is balanced across RWD treatment arms conditional on X . This is the standard no-unmeasured-confounding assumption in the RWD. Our framework deliberately avoids assuming it; the role of $c(x)$ is precisely to absorb the bias when this independence fails.

Implications for modelling and sensitivity. The structural form has two implications for what follows. First, $c(x)$ is small when prognostic imbalance is mild and effect modification by U is weak. The shrinkage prior on c [**PLACEHOLDER: to be introduced in the modelling section**] is then well-justified. Second, the magnitude of $c(x)$ admits a sensitivity-analysis interpretation: bounds on how strongly U is associated with A in the RWD and with f_0 translate into bounds on $c(x)$. [**PLACEHOLDER: brief discussion of relevant sensitivity-analysis frameworks (e.g., Rosenbaum, E-values, Robins–Rotnitzky) and where they fit in the paper.**]

2.9 Supplementary Materials: Structural causal model interpretation of the confounding function (expanded)

Context and goal. The previous section interprets $c(x)$ through a structural outcome model. That model is silent on how the other variables in the problem are generated. It posits a functional form for $\log T(a)$ but leaves the mechanisms behind X , U , S , and A unspecified. The main-text assumptions, Assumptions 1–4, carry the remaining structure as conditional independencies.

We give here a fuller treatment via a structural causal model (SCM; Pearl, 2009; Peters et al., 2017). An SCM specifies a structural equation for every endogenous variable and a joint distribution over exogenous noises. The resulting framework is still very broad. It is also a sub-class of the regression model used for inference in the main text. The SCM has three advantages over the outcome-model treatment. First, it makes the data-generating mechanism explicit. Second, it lets us recover the main-text assumptions from a small set of structural conditions. Third, it provides a mechanism-level language for sensitivity analyses on $c(x)$.

The structural causal model. We posit the following SCM over the random vector (X, U, S, A, T) :

$$\begin{aligned} X &= f_X(N_X), \\ U &= f_U(X, N_U), \\ S &= f_S(X, N_S), \\ A &= f_A(X, U, S, N_A), \\ T &= f_T(X, U, A, N_T). \end{aligned} \tag{26}$$

The exogenous noises $(N_X, N_U, N_S, N_A, N_T)$ are mutually independent. The structural functions $f_X, \dots, f_T$ are otherwise unrestricted. We model the source S as a function of X alone, so that N_U and N_S are independent. The latent vector U enters every downstream equation, reflecting that we leave its causal role open. The potential outcomes are obtained by intervention on A :

$$T(a) := f_T(X, U, a, N_T), \quad a \in \{0, 1\}. \tag{27}$$

The observational outcome equals $T(A)$ by construction, which recovers consistency.

We further restrict f_T to admit an additive separation of the baseline and the treatment effect on the log-time scale:

$$\log T(a) = f_0(X, U) + a \tau^*(X, U) + \varepsilon, \tag{28}$$

where $\varepsilon = \varepsilon(N_T)$ has mean zero and is independent of (X, U, A, S) . Equation (28) recovers the outcome model of the previous section. The remaining equations in (26) are new. They expose the selection mechanism f_S and the treatment-assignment mechanism f_A that the outcome model left implicit.

Graphical representation. The SCM induces the directed acyclic graph in Figure 1. The graph displays the causal mechanisms and makes the role of U visible. The arrow $U \rightarrow A$ is the source of confounding in the RWD. The same arrow is degenerate in the RCT under the randomisation restriction below.

Recovering the main-text assumptions. Each of the main-text assumptions maps to an explicit constraint on the SCM. We list them in turn.

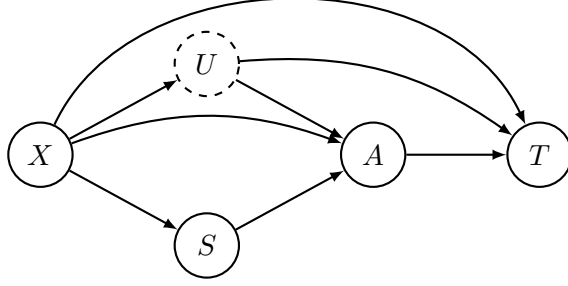


Figure 1: DAG implied by the SCM (26). Observed variables are solid; the unmeasured confounder U is dashed. The arrow $U \rightarrow A$ is degenerate when $S = 1$ under the RCT randomisation restriction (29).

Consistency (Assumption 1) holds by construction. The structural equation for T gives $T = f_T(X, U, A, N_T) = T(A)$.

Unconfoundedness of the RCT (Assumption 2) restricts f_A on the event $\{S = 1\}$. The RCT mechanism does not depend on U :

$$f_A(X, U, 1, N_A) = f_A^{\text{RCT}}(X, N_A). \quad (29)$$

Together with the independence of N_A from (N_U, N_T) , this restriction gives $A \perp\!\!\!\perp (U, T(0), T(1)) \mid X, S = 1$.

Positivity (Assumption 3) requires that the conditional distribution of A given (X, S) have full support on $\{0, 1\}$ in both sources. This is a non-degeneracy condition on f_A jointly with the noise N_A .

Cross-source exchangeability (Assumption 4) reduces to two conditions. First, f_T has no S argument, so the conditional distribution of $T(a)$ given (X, U) does not depend on S . Second, the distribution of U given X is the same in both sources:

$$U \mid X, S = 1 \stackrel{d}{=} U \mid X, S = 0. \quad (30)$$

The second condition follows from our construction. We let S depend on X but not on N_U , so $U \perp\!\!\!\perp S \mid X$. In the selection-diagram language of Bareinboim and Pearl (2016), S acts as a selection node attached to X but disconnected from U .

Conditionally independent censoring (Assumption 5) is a statement about the censoring mechanism. It is separate from the outcome SCM and we treat it as outside (26).

The SCM as a sub-class of the regression model. The regression model (7) places nonparametric priors on the conditional-mean components m_0 , τ , and c . It is agnostic about the joint distribution of (X, U, A, S) . The SCM (26)–(28) is one particular family of data-generating processes compatible with that regression model. Each component of the regression model arises as a conditional expectation under the SCM:

$$m_0(x, s) = \mathbb{E}[f_0(X, U) \mid X = x, S = s], \quad \tau(x) = \mathbb{E}[\tau^*(X, U) \mid X = x], \quad (31)$$

and $c(x)$ is the residual contrast defined by (4). The SCM family is still very broad. It permits arbitrary nonlinear mechanisms, multivariate U of unrestricted dimension, and arbitrary dependence of A on (X, U) in the RWD. The only structural restrictions are the additive form

(28), the source-symmetric outcome equation, and the independence (30) of U from S given X . The first restriction is the AFT modelling choice. The latter two are precisely what cross-source exchangeability buys us.

Identification within the SCM. The identification result of Proposition 2 carries over without change. The RCT slice of the SCM gives $\mathbb{E}[\log T \mid X, A = 1, S = 1] - \mathbb{E}[\log T \mid X, A = 0, S = 1] = \tau(X)$ via (29) and (30). The RWD slice identifies $\tau(x) + c(x)$ as the observed treated-versus-control contrast. The SCM contributes no additional identifying restrictions. The latent vector U remains unidentified.

Structural form of $c(x)$. We re-derive the structural form of the confounding function within the SCM. The argument is the same as in Proposition 3. The SCM makes the conditioning sets transparent. Decompositions of this form go back to Brumback et al. (2004) for marginal structural models and VanderWeele and Arah (2011) for general outcomes.

Proposition 4 (Structural form of c under the SCM). *Under the SCM (26)–(28) with restrictions (29) and (30), the confounding function satisfies*

$$c(x) = \Delta f_0(x) + \delta_\tau(x), \quad (32)$$

with the prognostic-imbalance term

$$\Delta f_0(x) = \mathbb{E}[f_0(X, U) \mid X = x, A = 1, S = 0] - \mathbb{E}[f_0(X, U) \mid X = x, A = 0, S = 0], \quad (33)$$

and the effect-modification term $\delta_\tau(x)$ defined as in (21).

Proof. We use (28) and the mean-zero independence of ε to write

$$\begin{aligned} \mathbb{E}[\log T \mid X=x, A=1, S=0] &= \mathbb{E}[f_0(X, U) + \tau^*(X, U) \mid X=x, A=1, S=0], \\ \mathbb{E}[\log T \mid X=x, A=0, S=0] &= \mathbb{E}[f_0(X, U) \mid X=x, A=0, S=0]. \end{aligned}$$

Subtraction gives the RWD treated-versus-control contrast as a sum of two terms in f_0 and one term in τ^* . We subtract $\tau(x) = \mathbb{E}[\tau^*(X, U) \mid X = x]$ on both sides. The definition (4) of $c(x)$ yields (32). $\square$

Structural conditions for $c \equiv 0$. The SCM lets us read off transparent sufficient conditions under which the confounding function vanishes.

Corollary 2 (Structural conditions for vanishing c). *Under the SCM, $c(x) \equiv 0$ if either of the following holds.*

- (i) *The RWD treatment mechanism does not depend on U : $f_A(X, U, 0, N_A) = f_A^{\text{RWD}}(X, N_A)$.*
- (ii) *Both f_0 and τ^* depend on covariates only through X .*

Either condition is sufficient. Neither is necessary.

Proof. Under (i), $U \perp\!\!\!\perp A \mid X, S = 0$. The conditional distribution of U given $(X, A, S = 0)$ does not depend on A . Both $\Delta f_0(x)$ and $\delta_\tau(x)$ vanish. Under (ii), $f_0(X, U) = f_0(X)$ and $\tau^*(X, U) = \tau(X)$. The conditional expectations in (32) reduce to $f_0(x)$ and $\tau(x)$, so $\Delta f_0 \equiv 0$ and $\delta_\tau \equiv 0$. $\square$

Condition (i) is the standard no-unmeasured-confounding assumption applied to the RWD. Our framework does not impose it. The confounding function exists precisely to absorb the bias when (i) fails. Condition (ii) describes a setting in which U is causally inert for the outcome. In that case the latent U can be marginalised away without loss.

Parametric instance. A linear-Gaussian instance of the SCM makes the structural sources of $c(x)$ tangible. Take $f_0(X, U) = \beta_0 + \beta_X^\top X + \beta_U U$ and $\tau^*(X, U) = \gamma_0 + \gamma_X^\top X + \gamma_U U$. Take an RWD treatment mechanism $\mathbb{P}(A = 1 \mid X, U, S = 0) = \Phi(\alpha_0 + \alpha_X^\top X + \alpha_U U)$, with $U \mid X \sim N(0, \sigma_U^2)$. The two pieces of (32) become

$$\Delta f_0(x) = \beta_U \{ \mathbb{E}[U \mid X=x, A=1, S=0] - \mathbb{E}[U \mid X=x, A=0, S=0] \}, \quad (34)$$

and

$$\delta_\tau(x) = \gamma_U \{ \mathbb{E}[U \mid X=x, A=1, S=0] - \mathbb{E}[U \mid X=x] \}. \quad (35)$$

Each conditional mean of U in the RWD is a monotone function of α_U . The size of $c(x)$ is governed by three structural quantities. The first is α_U , the strength of confounding in the treatment mechanism. The second is β_U , the baseline dependence on U . The third is γ_U , the strength of effect modification by U . Any one of them being zero kills the corresponding piece. All three zero recovers $c(x) \equiv 0$.

Sensitivity-analysis framing. The SCM provides mechanism-level handles for sensitivity analysis. Two structural quantities govern the magnitude of c . The first is the dependence of A on U in the RWD, governed by $f_A(\cdot, \cdot, 0, \cdot)$. The second is the dependence of f_0 and τ^* on U . Standard sensitivity-analysis frameworks parameterise these quantities and induce bounds on the bias. Rosenbaum bounds (Rosenbaum, 2002) constrain the odds-ratio dependence of A on U . The E-value of VanderWeele and Ding (2017) summarises the joint association strength required to nullify an observed effect. The framework of Robins et al. (2000) bounds the bias under prescribed deviations from ignorability. Yang et al. (2025) apply confounding-function methods to a setting close to ours. Each framework translates into an implied envelope on $c(x)$ via (32). The Bayesian counterpart of these analyses is the shrinkage prior on c introduced in the modelling section. The prior downweights large values of c without ruling them out. The data are free to pull c away from zero where the bias is strong.

Implications for modelling. The SCM clarifies what the shrinkage prior on c encodes. Shrinkage towards zero is consistent with the structural picture in which U is approximately balanced across RWD arms at most values of X and effect modification by U is mild. The prior does not force these conditions. The shrinkage strength controls how aggressively the RWD informs τ through the shared baseline m_0 . A weaker prior on c trusts the RWD less. A stronger prior on c approximates the conventional no-unmeasured-confounding assumption in the RWD. The full SCM treatment makes the prior interpretable as a soft version of Corollary 2(i). Related RCT–RWD fusion strategies that learn a confounding-bias function from a small experimental sample include Kallus et al. (2018) and Yang et al. (2025).

Supplementary Materials: Posterior computation for the HDPM error model

This section of the Supplementary Materials gives the full hierarchical model after truncation, the conditional distributions of each block of the Gibbs sampler, and recommendations for hyperparameter calibration and practical implementation.

Truncation and full hierarchical model. For computation we truncate the top-level stick-breaking at a finite level K (Ishwaran and James, 2001): $u_K = 1$, so $\beta_K = 1 - \sum_{k < K} \beta_k$ and $\beta_k = 0$ for $k > K$. Conditional on the truncated $\beta \in \Delta^{K-1}$ we use the finite-dimensional approximation:

$$\pi_s \mid \beta, M_s \sim \text{Dirichlet}(M_s \beta_1, \dots, M_s \beta_K), \quad (36)$$

the standard truncated representation of the hierarchical Dirichlet process (Teh et al., 2006; Wang et al., 2009). The approximation becomes exact as $K \rightarrow \infty$. We use a default $K = 50$, which has been adequate in our simulations. The largest occupied component index should be monitored across iterations to confirm that the truncation does not bind.

We introduce latent cluster assignments $Z_i \in \{1, \dots, K\}$ such that the latent location for observation i is $\theta_{Z_i}^* - \mu_{S_i}$. The full hierarchical model is:

$$\log T_i \mid Z_i, \theta^*, \sigma^2, m_0, \tau, c \sim N(\eta_i, \sigma_{S_i}^2), \quad (37)$$

$$\eta_i = m_0(X_i, S_i) + A_i \tau(X_i) + A_i(1 - S_i) c(X_i) + \theta_{Z_i}^* - \mu_{S_i}, \quad (38)$$

$$Z_i \mid \pi_{S_i} \sim \text{Categorical}(\pi_{S_i}), \quad (39)$$

$$\pi_s \mid \beta, M_s \sim \text{Dirichlet}(M_s \beta_1, \dots, M_s \beta_K), \quad s \in \{0, 1\}, \quad (40)$$

$$\beta \mid \gamma \sim \text{Stick}_K(\gamma), \quad (41)$$

$$\theta_k^* \stackrel{\text{iid}}{\sim} N(0, \sigma_\theta^2), \quad k = 1, \dots, K, \quad (42)$$

$$m_0, \tau, c \sim \text{BART}, \quad (43)$$

$$\sigma_s^2 \sim \nu \lambda / \chi_\nu^2, \quad s \in \{0, 1\}, \quad (44)$$

$$\gamma \sim \text{Gamma}(a_\gamma, b_\gamma), \quad (45)$$

$$M_s \sim \text{Gamma}(a_M, b_M), \quad s \in \{0, 1\}, \quad (46)$$

with $\mu_s = \sum_k \pi_{sk} \theta_k^*$ a deterministic function of π_s and θ^* . The notation $\text{Stick}_K(\gamma)$ denotes the truncated stick-breaking distribution with $\text{Beta}(1, \gamma)$ breaks.

Gibbs sampler. For each observation i , define the partial residual:

$$R_i := \log T_i - m_0(X_i, S_i) - A_i \tau(X_i) - A_i(1 - S_i) c(X_i). \quad (47)$$

Conditional on the BART components and the latent cluster assignment $Z_i = k$, we have $R_i \sim N(\theta_k^* - \mu_{S_i}, \sigma_{S_i}^2)$. One sweep of the sampler executes the following steps.

Step 1 (data augmentation for censoring). For each i with $\delta_i = 0$, draw:

$$\log T_i \mid \cdot \sim \text{TruncatedNormal}(\eta_i, \sigma_{S_i}^2; [\log C_i, \infty)), \quad (48)$$

with η_i as defined above.

Step 2 (BART backfitting). UPDATE THIS SECTION! We have four forests, also the deviation forest. The three forests are updated sequentially via Bayesian backfitting (Chipman et al., 2010). The working response for each forest is the partial residual obtained by subtracting all other components and the current cluster shift $\theta_{Z_i}^* - \mu_{S_i}$:

$$R_i^{(m_0)} = \log T_i - A_i \tau(X_i) - A_i(1 - S_i)c(X_i) - (\theta_{Z_i}^* - \mu_{S_i}), \quad (49)$$

$$R_i^{(\tau)} = \log T_i - m_0(X_i, S_i) - (1 - S_i)c(X_i) - (\theta_{Z_i}^* - \mu_{S_i}), \quad i \text{ with } A_i = 1, \quad (50)$$

$$R_i^{(c)} = \log T_i - m_0(X_i, 0) - \tau(X_i) - (\theta_{Z_i}^* - \mu_0), \quad i \text{ with } A_i = 1, S_i = 0. \quad (51)$$

Each working response is treated as Gaussian with mean equal to the relevant forest and per-observation variance $\sigma_{S_i}^2$. Control observations carry no information about τ . RCT observations and untreated RWD observations carry no information about c . After the BART updates, recompute R_i from (47).

Step 3 (cluster assignments). For each i , sample:

$$P(Z_i = k \mid \cdot) \propto \pi_{S_i, k} \phi\left(\frac{R_i - (\theta_k^* - \mu_{S_i})}{\sigma_{S_i}}\right), \quad k = 1, \dots, K. \quad (52)$$

Step 4 (unconstrained atoms). Atoms are sampled in the unconstrained parameterisation and the centring is reapplied deterministically. Let $n_{sk} = \sum_i \mathbb{1}\{S_i = s, Z_i = k\}$ and $\tilde{R}_{sk} = \sum_{i: Z_i=k, S_i=s} (R_i + \mu_s)$. The conjugate Gaussian update is precision-weighted across sources:

$$\theta_k^* \mid \cdot \sim N\left(\frac{\sum_s \sigma_s^{-2} \tilde{R}_{sk}}{\sigma_\theta^{-2} + \sum_s \sigma_s^{-2} n_{sk}}, (\sigma_\theta^{-2} + \sum_s \sigma_s^{-2} n_{sk})^{-1}\right), \quad k = 1, \dots, K. \quad (53)$$

After updating all atoms, recompute $\mu_s = \sum_k \pi_{sk} \theta_k^*$ and $\theta_{sk} = \theta_k^* - \mu_s$ for $s \in \{0, 1\}$. As discussed in Yang et al. (2010), this parameter-expansion strategy preserves the correct marginal posterior for the centred quantities.

Step 5 (top-level sticks). The top-level stick weights are updated using the auxiliary table-count augmentation of Teh et al. (2006). Let $n_{sk} = \sum_i \mathbb{1}\{S_i = s, Z_i = k\}$. For each (s, k) with $n_{sk} > 0$, sample the auxiliary table count t_{sk} via the Antoniak representation (Antoniak, 1974):

$$t_{sk} = \sum_{j=1}^{n_{sk}} B_{sk,j}, \quad B_{sk,j} \sim \text{Bernoulli}\left(\frac{M_s \beta_k}{M_s \beta_k + j - 1}\right), \quad (54)$$

and set $t_{sk} = 0$ if $n_{sk} = 0$. Let $t_{\cdot k} = t_{0k} + t_{1k}$. Update the truncated sticks:

$$u_k \mid \cdot \sim \text{Beta}\left(1 + t_{\cdot k}, \gamma + \sum_{l>k} t_{\cdot l}\right), \quad k = 1, \dots, K-1, \quad (55)$$

with $u_K = 1$, and recompute $\beta_k = u_k \prod_{l<k} (1 - u_l)$.

Step 6 (source-specific weights). For each $s \in \{0, 1\}$, sample:

$$\pi_s \mid \cdot \sim \text{Dirichlet}(M_s \beta_1 + n_{s1}, \dots, M_s \beta_K + n_{sK}). \quad (56)$$

Recompute μ_s and the centred atoms.

Step 7 (concentration parameters). The concentration parameters γ and M_0, M_1 are updated using the auxiliary-variable approach of [Escobar and West \(1995\)](#). For γ , let K^* denote the number of occupied top-level components and $t_{..} = \sum_{s,k} t_{sk}$ the total table count. Sample $\eta_\gamma \mid \gamma, t_{..} \sim \text{Beta}(\gamma + 1, t_{..})$, then:

$$\gamma \mid \cdot \sim \omega_\gamma \text{Gamma}(a_\gamma + K^*, b_\gamma - \log \eta_\gamma) + (1 - \omega_\gamma) \text{Gamma}(a_\gamma + K^* - 1, b_\gamma - \log \eta_\gamma), \quad (57)$$

with $\omega_\gamma / (1 - \omega_\gamma) = (a_\gamma + K^* - 1) / [t_{..}(b_\gamma - \log \eta_\gamma)]$. The analogous update for M_s replaces K^* by the source-specific table count $t_{s\cdot} = \sum_k t_{sk}$ and uses $\eta_{M,s} \sim \text{Beta}(M_s + 1, n_s)$. See [Teh et al. \(2006, Appendix A\)](#) for the derivation.

Step 8 (error scale). The conjugate update for the residual variance is:

$$\sigma_s^2 \mid \cdot \sim \text{InverseGamma}\left(\frac{\nu + n_s}{2}, \frac{\nu\lambda + \hat{s}_s^2}{2}\right), \quad \hat{s}_s^2 = \sum_{i:S_i=s} (R_i - (\theta_{Z_i}^* - \mu_s))^2, \quad s \in \{0, 1\}. \quad (58)$$

Hyperparameter calibration. We adopt the BART defaults of [Chipman et al. \(2010\)](#) with the AFT-specific adaptations of [Henderson et al. \(2020\)](#): $J = 200$ trees, $\alpha = 0.95$, $\beta = 2$, and the response transformation $y_i^{\text{tr}} = y_i \exp(-\hat{\mu}_{\text{AFT}})$ based on a preliminary parametric log-normal AFT fit. The error-scale hyperparameters λ and σ_θ^2 are calibrated using the Yamato-type approximation ([Yamato, 1984](#)). The quantity $\sum_k \pi_{sk} (\theta_k^* - \mu_s)^2 / \sigma_\theta^2$ is approximately χ_1^2 in distribution. The induced prior on the marginal residual variance $\text{Var}(\varepsilon \mid S = s)$ can therefore be matched in tail probability to a preliminary parametric estimate $\hat{\sigma}_\varepsilon^2$. The approximation applies within each source separately. We use a default tail probability $q = 0.5$.

The concentration hyperpriors are $\gamma \sim \text{Gamma}(a_\gamma, b_\gamma)$ and $M_s \sim \text{Gamma}(a_M, b_M)$ for $s \in \{0, 1\}$, with $a_\gamma = a_M = 2$ and $b_\gamma = b_M = 0.1$. This places prior mean 20 on each concentration with prior mode 10.

Practical considerations. Track $\max_i Z_i$ across iterations. If the largest occupied index equals K in more than a negligible fraction of iterations, the truncation is too tight and K should be increased. For initialisation, use the standard [Chipman et al. \(2010\)](#) initialisation for the BART components. Cluster labels can be assigned by k -means on residuals from a preliminary parametric fit. Set $\beta = (1/K, \dots, 1/K)$, $\pi_s = \beta$, and $\gamma = M_0 = M_1 = 1$.

Supplementary Materials: The hierarchical m_0 prior as a meta-analytic-predictive prior

The meta-analytic-predictive (MAP) prior of [Neuenschwander et al. \(2010\)](#) is a hierarchical Bayesian construction for borrowing strength between studies. We derive our specification (??) of $m_0(X, S)$ as a function-valued instance of the MAP construction.

Setup of the MAP prior. Consider K studies with study-specific parameters $\theta_1, \dots, \theta_K$. The MAP prior assumes a Gaussian hierarchical structure:

$$\theta_k \mid \mu, \sigma_d^2 \sim N(\mu, \sigma_d^2), \quad k = 1, \dots, K, \quad (59)$$

$$\mu \sim p(\mu), \quad (60)$$

$$\sigma_d \sim p(\sigma_d). \quad (61)$$

The shared mean μ is the borrowing target. The heterogeneity variance σ_d^2 controls how strongly each θ_k is shrunk towards μ . Small σ_d enforces near-equality of the θ_k , i.e. strong borrowing. Large σ_d allows the θ_k to drift apart, i.e. weak borrowing. The hierarchical structure lets the data determine the degree of borrowing through the posterior of σ_d .

Function-valued extension. Our two studies are the RWD ($s = 0$) and the RCT ($s = 1$), and the study-specific parameter is the prognostic function evaluated at source s :

$$\theta_s(X) := m_0(X, s), \quad s \in \{0, 1\}. \quad (62)$$

We extend (61) from a scalar to a function by replacing the shared mean with a shared function and the Gaussian deviation with a function-valued deviation:

$$m_0(X, s) = m_0^{\text{sh}}(X) + d_s(X), \quad s \in \{0, 1\}, \quad (63)$$

$$m_0^{\text{sh}} \sim \text{BART}(\sigma_{h, \text{sh}}^2), \quad (64)$$

$$d_s \sim \text{BART}(\sigma_d^2), \quad \text{independently across } s. \quad (65)$$

The BART prior plays the role of a nonparametric prior over functions. The leaf-prior variance σ_d^2 now controls cross-source heterogeneity in the same way as the scalar MAP. The construction is heterogeneous because the deviation is allowed to vary with X . The standard MAP assigns a hyperprior to σ_d and lets the data drive borrowing through its posterior. We instead treat σ_d as a calibrated hyperparameter, in line with standard BART practice for leaf-prior scales; the calibration is given below.

Anchored simplification used in the main text. The main-text specification (??) anchors at the RCT by setting $d_1 \equiv 0$ and writing $d_0 = d$, giving:

$$m_0(X, S) = m_0^{\text{sh}}(X) + (1 - S) \cdot d(X). \quad (66)$$

This anchors the prognostic function at the RCT level and treats the RWD as the source that drifts from it. The anchored form is parsimonious and integrates cleanly with the backfitting sampler. The fully symmetric form (65) with separate d_0 and d_1 retains the same identifiability of τ and c and is preferable when neither source plays the role of a natural reference.

Semiparametric extension of Hupf et al. (2021). The standard MAP (61) assumes Gaussian deviations, which is restrictive when one or more studies are outliers. Hupf et al. (2021) extend MAP to flexible deviations through a Dirichlet-process-based prior on the heterogeneity distribution. The BART-based deviation in our functional MAP (65) plays an analogous role at the function level. It allows the shape of d_s to depart from a parametric form whenever the data demand it, while retaining the borrowing target m_0^{sh} .

Calibration of the borrowing scale. We treat σ_d as a calibrated hyperparameter rather than sampling it. The leaf-prior standard deviation of the deviation forest is set by the Chipman–George–McCulloch rule (Chipman et al., 2010):

$$\sigma_d = \frac{k_d}{2\sqrt{m_d}}, \quad (67)$$

where m_d is the number of trees in the deviation forest and $k_d > 0$ is the tuning constant of the Chipman calibration applied to the deviation forest, playing the role of the borrowing-strength dial. Small k_d shrinks d towards zero and induces strong borrowing of the RWD prognostic towards the RCT prognostic. Large k_d relaxes the prior and lets d depart further from zero. We take $k_d = 1$ by default, in line with the uniform Chipman calibration of the main text. A fully Bayesian alternative is to put a hyperprior on σ_d as in the standard MAP (Neuenschwander et al., 2010); this trades the simplicity of the BART convention for additional data-driven adaptation of the borrowing strength.

Supplementary Materials: Outer Gibbs sampler and tree updates

The posterior is sampled by a blocked Gibbs sampler built on an extended Bayesian backfitting approach (Chipman et al., 2010; Hahn et al., 2020). One sweep of the sampler is summarised in Algorithm 1. The sampler interleaves data augmentation for censored observations, single-tree reversible-jump updates of the four forests, an update of the borrowing scale σ_d , and an update of the HDPM error block.

Single-tree reversible-jump updates. Each tree-and-step-height pair $(\mathcal{T}_j^f, \mathcal{H}_j^f)$ is updated by one of three reversible-jump proposals: GROW, PRUNE, or CHANGE. The proposal mechanism follows Chipman et al. (2010) with the AFT-specific adaptations of Henderson et al. (2020). Full derivations of the acceptance ratios are given in the references cited above and are not reproduced here. The conjugate Gaussian update of the leaf step heights is performed jointly with the structural proposal.

Partial residuals. Each forest is updated against a partial residual that subtracts the contributions of the other three forests and the current cluster shift $\theta_{Z_i}^* - \mu_{S_i}$ from the residual mixture. The active subset of observations differs across forests. The shared baseline m_0^{sh} uses all observations. The deviation d uses the RCT observations only. The treatment effect τ uses the treated observations from both sources. The confounding function c uses the treated RWD observations.

Borrowing-scale update. The borrowing scale σ_d of the heterogeneous MAP prior in Supplementary Materials 2.9 is sampled from its conjugate posterior at the end of each sweep.

HDPM error block. The blocked update of the error distribution is summarised in Algorithm 2. Each step corresponds to a conjugate Categorical, Gaussian, Beta, Dirichlet, Gamma, or inverse-Gamma update. The full mathematical specification of every step is given in Supplementary Materials 2.9.

Data augmentation for censoring. Observations with $\delta_i = 0$ have their unobserved $\log T_i$ drawn from a truncated normal restricted to $[\log C_i, \infty)$ at the start of each sweep (Tanner and Wong, 1987).

Algorithm 1: Outer Gibbs sampler — one sweep

Input: Current forests $(\mathcal{T}_j^f, \mathcal{H}_j^f)$ for $f \in \{m_0^{\text{sh}}, d, \tau, c\}$; observed follow-up times $\tilde{T}_i$ and event indicators δ_i ; HDPM error state; borrowing scale σ_d .

- 1 **for** $i = 1$ **to** n **with** $\delta_i = 0$ **do**
 - 2 $\lfloor$ Draw $\log T_i$ from a truncated normal restricted to $[\log C_i, \infty)$;
 - 3 **foreach** forest $f \in \{m_0^{\text{sh}}, d, \tau, c\}$ **do**
 - 4 **for** $j = 1$ **to** m_f **do**
 - 5 Compute the partial residual $\mathcal{R}_j^f$ on the active subset for f ;
 - 6 Update $(\mathcal{T}_j^f, \mathcal{H}_j^f)$ by a single-tree reversible-jump step;
 - 7 Update the borrowing scale σ_d from its conjugate posterior;
 - 8 Update the HDPM error block (Algorithm 2);
-

Supplementary Materials: Posterior computation *[merged draft for review]*

Merged and streamlined replacement of the previous Supplementary-Materials sections on the HDPM error model (2.9) and the outer Gibbs sampler (2.9). Added as a separate section so the originals can be compared against it before deletion.

These Supplementary Materials specify the full conditional distribution of every block of the posterior sampler. We sample the posterior by a blocked Gibbs sampler built on extended Bayesian backfitting (Hastie and Tibshirani, 2000). The next subsections describe the outer sampler at a high level and then give the update for each block in turn.

Outer Gibbs sampler

One iteration of the sampler consists of three blocks executed in order: updates of the four forests $f \in \{m_0^{\text{sh}}, d, \tau, c\}$ in turn, an update of the HDPM error block, and data augmentation of the censored event times. Algorithm 3 summarises one iteration, and Algorithm 4 summarises the inner update of the HDPM error block. After burn-in, each iteration produces one posterior sample of the full parameter vector. The leaf-variance hyperparameters of the four forests, including the borrowing scale σ_d of the deviation forest, are treated as calibrated hyperparameters and held fixed across iterations. Their calibration is described at the end of these Supplementary Materials.

Forest updates

The four forests are updated sequentially by Bayesian backfitting. Each tree is updated by a birth-death Metropolis-Hastings step on the tree structure, following Chipman et al. (2010) with the AFT-specific adaptations of Henderson et al. (2020). The leaf step heights are drawn from their conjugate Gaussian posterior conditional on the tree structure. Each forest is updated against a partial residual that subtracts the contributions of the other three forests and the current cluster shift $\theta_{Z_i}^* - \mu_{S_i}$ from $\log T_i$. The active subset of observations differs across forests: m_0^{sh} uses all observations, d uses the RCT observations, τ uses the treated observations from both sources, and c uses the treated RWD observations.

Algorithm 2: HDPM error block — one update

Input: Partial residuals $R_i = \log T_i - m_0^{\text{sh}}(X_i) - S_i d(X_i) - A_i \tau(X_i) - A_i(1 - S_i)c(X_i)$;
current atoms θ^* , sticks β , source weights π_0, π_1 , concentrations γ, M_0, M_1 ,
error scales σ_0^2, σ_1^2 .

- 1 **for** $i = 1$ **to** n **do**
 - 2 | Sample cluster assignment Z_i from the categorical full conditional;
 - 3 **for** $k = 1$ **to** K **do**
 - 4 | Update the unconstrained atom θ_k^* from its conjugate Gaussian posterior;
 - 5 Recompute $\mu_s = \sum_k \pi_{sk} \theta_k^*$ and the centred atoms;
 - 6 Sample auxiliary table counts t_{sk} via the Antoniak representation;
 - 7 Update top-level sticks u_k from Beta posteriors, then recompute β_k ;
 - 8 **foreach** $s \in \{0, 1\}$ **do**
 - 9 | Update source weights π_s from a Dirichlet posterior;
 - 10 Recompute μ_s and the centred atoms;
 - 11 Update concentrations γ, M_0, M_1 via the Escobar–West auxiliary-variable scheme;
 - 12 **foreach** $s \in \{0, 1\}$ **do**
 - 13 | Update the error scale σ_s^2 from its conjugate inverse-gamma posterior;
-

Updating the HDPM error block

The HDPM error block updates the latent variables of the nonparametric error distribution: cluster assignments, atoms, top-level stick weights, source-specific weights, concentration parameters, and the per-source residual variances. All full conditionals are conjugate. We first state the truncation of the top-level stick-breaking and the resulting hierarchical model, and then give the full conditional for each update in turn.

For computation we truncate the top-level stick-breaking at a finite level K (Ishwaran and James, 2001), setting $u_K = 1$ so that $\beta_K = 1 - \sum_{k < K} \beta_k$ and $\beta_k = 0$ for $k > K$. Conditional on the truncated $\beta \in \Delta^{K-1}$, the source-specific weights have a finite-dimensional Dirichlet prior:

$$\pi_s \mid \beta, M_s \sim \text{Dirichlet}(M_s \beta_1, \dots, M_s \beta_K), \quad (68)$$

the standard truncated representation of the hierarchical Dirichlet process (Teh et al., 2006; Wang et al., 2009). The approximation becomes exact as $K \rightarrow \infty$, and we use a default of $K = 50$. We introduce latent cluster assignments $Z_i \in \{1, \dots, K\}$ such that the latent location for observation i is $\theta_{Z_i}^* - \mu_{S_i}$, with $\mu_s = \sum_k \pi_{sk} \theta_k^*$. The full hierarchical model after truncation is then:

$$\log T_i \mid \cdot \sim N(\eta_i, \sigma_{S_i}^2), \quad (69)$$

$$Z_i \mid \pi_{S_i} \sim \text{Categorical}(\pi_{S_i}), \quad (70)$$

$$\pi_s \mid \beta, M_s \sim \text{Dirichlet}(M_s \beta_1, \dots, M_s \beta_K), \quad s \in \{0, 1\}, \quad (71)$$

$$\beta \mid \gamma \sim \text{Stick}_K(\gamma), \quad (72)$$

$$\theta_k^* \stackrel{\text{iid}}{\sim} N(0, \sigma_\theta^2), \quad k = 1, \dots, K, \quad (73)$$

$$\sigma_s^2 \sim \nu \lambda / \chi_\nu^2, \quad s \in \{0, 1\}, \quad (74)$$

$$\gamma \sim \text{Gamma}(a_\gamma, b_\gamma), \quad (75)$$

$$M_s \sim \text{Gamma}(a_M, b_M), \quad s \in \{0, 1\}, \quad (76)$$

where $\text{Stick}_K(\gamma)$ denotes the truncated stick-breaking distribution with $\text{Beta}(1, \gamma)$ breaks.

The full conditionals below are all conjugate given the forests and the augmented event times. We define the partial residual

$$R_i := \log T_i - m_0^{\text{sh}}(X_i) - (1 - S_i) d(X_i) - A_i \tau(X_i) - A_i(1 - S_i) c(X_i), \quad (77)$$

so that $R_i \mid Z_i = k \sim N(\theta_k^* - \mu_{S_i}, \sigma_{S_i}^2)$. The cluster assignments are then sampled by

$$P(Z_i = k \mid \cdot) \propto \pi_{S_i, k} \phi\left(\frac{R_i - (\theta_k^* - \mu_{S_i})}{\sigma_{S_i}}\right), \quad k = 1, \dots, K. \quad (78)$$

We sample atoms in the unconstrained parameterisation and reapply centring deterministically. Let $n_{sk} = \sum_i \mathbb{1}\{S_i = s, Z_i = k\}$ and $\tilde{R}_{sk} = \sum_{i: Z_i=k, S_i=s} (R_i + \mu_s)$. The conjugate Gaussian update is precision-weighted across sources:

$$\theta_k^* \mid \cdot \sim N\left(\frac{\sum_s \sigma_s^{-2} \tilde{R}_{sk}}{\sigma_\theta^{-2} + \sum_s \sigma_s^{-2} n_{sk}}, (\sigma_\theta^{-2} + \sum_s \sigma_s^{-2} n_{sk})^{-1}\right), \quad k = 1, \dots, K. \quad (79)$$

We then recompute $\mu_s = \sum_k \pi_{sk} \theta_k^*$ and the centred atoms $\theta_{sk} = \theta_k^* - \mu_s$. This parameter-expansion strategy preserves the correct marginal posterior for the centred quantities (Yang et al., 2010).

The top-level stick weights are updated by the auxiliary table-count augmentation of Teh et al. (2006). For each (s, k) with $n_{sk} > 0$ we sample the auxiliary table count via the Antoniak representation (Antoniak, 1974):

$$t_{sk} = \sum_{j=1}^{n_{sk}} B_{sk,j}, \quad B_{sk,j} \sim \text{Bernoulli}\left(\frac{M_s \beta_k}{M_s \beta_k + j - 1}\right), \quad (80)$$

and set $t_{sk} = 0$ when $n_{sk} = 0$. With $t_{\cdot k} = t_{0k} + t_{1k}$, the truncated sticks are updated by

$$u_k \mid \cdot \sim \text{Beta}\left(1 + t_{\cdot k}, \gamma + \sum_{l>k} t_{\cdot l}\right), \quad k = 1, \dots, K-1, \quad (81)$$

with $u_K = 1$ and $\beta_k = u_k \prod_{l<k} (1 - u_l)$. The source-specific weights then follow

$$\pi_s \mid \cdot \sim \text{Dirichlet}(M_s \beta_1 + n_{s1}, \dots, M_s \beta_K + n_{sK}), \quad s \in \{0, 1\}, \quad (82)$$

after which we recompute μ_s and the centred atoms.

The concentration parameters γ and M_0, M_1 are updated by the auxiliary-variable scheme of Escobar and West (1995). Let K^* denote the number of occupied top-level components and $t_{\cdot} = \sum_{s,k} t_{sk}$ the total table count. We draw $\eta_\gamma \mid \gamma, t_{\cdot} \sim \text{Beta}(\gamma + 1, t_{\cdot})$ and then

$$\gamma \mid \cdot \sim \omega_\gamma \text{Gamma}(a_\gamma + K^*, b_\gamma - \log \eta_\gamma) + (1 - \omega_\gamma) \text{Gamma}(a_\gamma + K^* - 1, b_\gamma - \log \eta_\gamma), \quad (83)$$

with $\omega_\gamma / (1 - \omega_\gamma) = (a_\gamma + K^* - 1) / [t_{\cdot} (b_\gamma - \log \eta_\gamma)]$. The analogous update for M_s replaces K^* by the source-specific table count $t_{s\cdot} = \sum_k t_{sk}$ and uses $\eta_{M,s} \sim \text{Beta}(M_s + 1, n_s)$. The derivation is in Teh et al. (2006, Appendix A).

The conjugate update of the residual variance closes the block:

$$\sigma_s^2 \mid \cdot \sim \text{InverseGamma}\left(\frac{\nu + n_s}{2}, \frac{\nu \lambda + \hat{s}_s^2}{2}\right), \quad \hat{s}_s^2 = \sum_{i: S_i=s} (R_i - (\theta_{Z_i}^* - \mu_s))^2, \quad s \in \{0, 1\}. \quad (84)$$

Algorithm 3: Outer Gibbs sampler, one iteration.

Input: Current forests $(\mathcal{T}_j^f, \mathcal{H}_j^f)$ for $f \in \{m_0^{\text{sh}}, d, \tau, c\}$; current augmented event times $\log T_i$; HDPM error state. Calibrated hyperparameters $\sigma_{h,f}$ for each forest are held fixed.

```

1 foreach forest  $f \in \{m_0^{\text{sh}}, d, \tau, c\}$  do
2   for  $j = 1$  to  $m_f$  do
3     Compute the partial residual  $\mathcal{R}_j^f$  on the active subset for  $f$ ;
4     Update  $\mathcal{T}_j^f$  by a birth-death Metropolis-Hastings step, then draw  $\mathcal{H}_j^f$  from its
       conjugate posterior;
5 Update the HDPM error block (Algorithm 4);
6 for  $i = 1$  to  $n$  with  $\delta_i = 0$  do
7   Draw  $\log T_i$  from a truncated normal restricted to  $[\log C_i, \infty)$ ;
```

Data augmentation for censored event times

Each iteration closes with data augmentation of the unobserved event times for right-censored observations (Tanner and Wong, 1987). For every i with $\delta_i = 0$ we draw

$$\log T_i \mid \cdot \sim \text{TruncatedNormal}(\eta_i, \sigma_{S_i}^2; [\log C_i, \infty)), \quad (85)$$

where $\eta_i = m_0^{\text{sh}}(X_i) + (1 - S_i)d(X_i) + A_i\tau(X_i) + A_i(1 - S_i)c(X_i) + \theta_{Z_i}^* - \mu_{S_i}$ is the conditional mean of $\log T_i$ at the current parameter values. The augmented values then serve as the working response for the next iteration.

Hyperparameter calibration and initialisation

We adopt the BART defaults of Chipman et al. (2010) with the AFT-specific adaptations of Henderson et al. (2020): $J = 200$ trees, $\alpha = 0.95$, $\beta = 2$, and the response transformation $y_i^{\text{tr}} = y_i \exp(-\hat{\mu}_{\text{AFT}})$ based on a preliminary parametric log-normal AFT fit. The leaf-prior standard deviation of each forest f is calibrated by the Chipman–George–McCulloch rule $\sigma_{h,f} = k_f / (2\sqrt{m_f})$, where m_f is the number of trees in the forest and $k_f > 0$ is a tuning constant. The borrowing scale of the deviation forest is the special case $\sigma_d = k_d / (2\sqrt{m_d})$. Small k_d shrinks the deviation d towards zero and induces strong borrowing of the RWD prognostic towards the RCT prognostic; large k_d relaxes the prior and lets d depart further from zero. The default $k_f = 1$ for every forest f matches the main-text calibration. The error-scale hyperparameters λ and σ_θ^2 are calibrated by a Yamato-type approximation (Yamato, 1984). The quantity $\sum_k \pi_{sk}(\theta_k^* - \mu_s)^2 / \sigma_\theta^2$ is approximately χ_1^2 , so the induced prior on the marginal residual variance $\text{Var}(\varepsilon \mid S = s)$ can be matched in tail probability to a preliminary parametric estimate $\hat{\sigma}_\varepsilon^2$. The approximation is applied within each source separately, with a default tail probability of $q = 0.5$. For the concentration hyperpriors we use $a_\gamma = a_M = 2$ and $b_\gamma = b_M = 0.1$, which places prior mean 20 and prior mode 10 on each concentration. We initialise the BART components by the defaults of Chipman et al. (2010), the cluster labels by k -means on residuals from a preliminary parametric fit, and the HDPM weights by $\beta = (1/K, \dots, 1/K)$, $\pi_s = \beta$, and $\gamma = M_0 = M_1 = 1$. We track $\max_i Z_i$ across iterations; if it equals K in more than a negligible fraction of iterations, the truncation level K should be increased.

Algorithm 4: Update of the HDPM parameters block.

Input: Partial residuals R_i from (77); current atoms θ^* , sticks β , source weights π_0, π_1 , concentrations γ, M_0, M_1 , error scales σ_0^2, σ_1^2 .

- 1 **for** $i = 1$ **to** n **do**
 - 2 $\lfloor$ Sample cluster assignment Z_i from the categorical full conditional in (78);
 - 3 **for** $k = 1$ **to** K **do**
 - 4 $\lfloor$ Update the unconstrained atom θ_k^* from its conjugate Gaussian posterior in (79);
 - 5 Recompute $\mu_s = \sum_k \pi_{sk} \theta_k^*$ and the centred atoms;
 - 6 Sample auxiliary table counts t_{sk} via the Antoniak representation;
 - 7 Update top-level sticks u_k from Beta posteriors, then recompute β_k ;
 - 8 **foreach** $s \in \{0, 1\}$ **do**
 - 9 $\lfloor$ Update source weights π_s from a Dirichlet posterior;
 - 10 Recompute μ_s and the centred atoms;
 - 11 Update concentrations γ, M_0, M_1 via the Escobar-West auxiliary-variable scheme;
 - 12 **foreach** $s \in \{0, 1\}$ **do**
 - 13 $\lfloor$ Update the error scale σ_s^2 from its conjugate inverse-gamma posterior;
-

Supplementary Materials: Section structure and writing status

This section records the current state of the methodology section: the proposed final structure, what is drafted, what remains, and the open items that need attention.

Proposed structure of the methodology section. *Main text.*

1. **Causal framework** — data sources, notation, potential outcomes, censoring formulation, CATE and acceleration factor, assumptions, and the identification trade-off (RWD unconfoundedness against CATE transportability).
2. **Accelerated failure time decomposition** — confounding function $c(x)$, the additive decomposition of $\mathbb{E}[\log T \mid A, X, S]$, identification of τ and c , the AFT outcome model.
3. **Nonparametric error distribution** — hierarchical Dirichlet process mixture of normals (HDPM) with source-specific scales σ_s .
4. **Overview of BART** — Bayesian additive regression trees as the prior for the regression functions, with the $\text{BART}(m, k, \alpha, \beta)$ shorthand and the tree-structure / step-height prior decomposition.
5. **Prior specification** — BART priors on $m_0^{\text{sh}}, d, \tau, c$ with regularisation matched to each component, the σ_s scaled-inverse- χ^2 prior, standardisation of the response, and the hierarchical model summary.
6. **Posterior inference** — hierarchical Bayesian Bootstrap for marginalising the CATE, posterior computation via extended Bayesian backfitting, and ideas for posterior summaries of heterogeneity, borrowing, and residual fit still to be developed.

Supplementary Materials.

- A. **Proofs** — Propositions 1 and 2.
- B. **Structural interpretation of c** — reduced-form structural reading via an unmeasured confounder U .
- C. **Structural causal model for c (expanded)** — full SCM, DAG, parametric instance, sensitivity-analysis framing.
- D. **Posterior computation for the HDPM error model** — truncation, full hierarchical model, eight-step Gibbs sampler with source-specific scales, hyperparameter calibration.
- E. **Hierarchical m_0 as a meta-analytic-predictive prior** — derivation from the MAP framework of Neuenschwander et al. (2010).
- F. **Outer Gibbs sampler and tree updates** — extended Bayesian backfitting algorithm and HDPM error block, with two algorithm boxes.

Status overview.

Section	Status	Notes
<i>Main text.</i>		
1. Causal framework	Drafted	Interval-censoring formulation in place; identification trade-off paragraph proofread.
2. AFT decomposition	Drafted	Propositions, proofs, structural-reading remark; meta-comment on the m_0 parametrisation still embedded.
3. Nonparametric error distribution	Drafted	HDPM construction with per-source σ_s and the $\sigma_s^2 \sim \nu\lambda/\chi_\nu^2$ prior calibrated following Chipman et al. (2010) .
4. Overview of BART	Drafted	Concise; tree-structure and step-height prior named explicitly.
5. Prior specification	Drafted	Regularisation framing matched to each component; recommended default $k_f = 1$ across the four forests, supported by the prior simulation study.
6. Posterior inference	Partially drafted	Hierarchical Bayesian Bootstrap for the ATE drafted; posterior-computation paragraph drafted; itemised placeholders for treatment-effect summaries, heterogeneity, borrowing, and residual fit still to be filled out.
<i>Supplementary Materials.</i>		
A. Proofs	Drafted	Two short proofs.
B. Structural interpretation of c	Drafted	Reduced-form reading with proof.
C. Structural causal model (expanded)	Drafted	Overlap with B; decide whether to merge or to clearly delineate roles.
D. Posterior computation HDPM	Drafted	Eight-step sampler with source-specific scales; atom update is precision-weighted.
E. Hierarchical m_0 MAP prior	Drafted	Functional MAP derivation; half-Normal scale hyperprior placeholder.
F. Outer Gibbs sampler and tree updates	Drafted	Algorithm 1 (outer sweep) and Algorithm 2 (HDPM error block).
G. Posterior computation [merged draft]	For-review only	Streamlined merger of D and F kept alongside the originals; pending decision before deletion of D + F or of G.

Outstanding items.

- **Section 6 (Posterior inference) is partially drafted.** The ATE/BB paragraph and the posterior-computation paragraph are in place. The four itemised topics (treatment-effect estimands, heterogeneity, borrowing, residual fit) are still placeholder bullets and need to be expanded into prose.
- **Inline [PLACEHOLDER: ...] markers** remain at two locations in the SCM Supplementary Materials: the shrinkage-prior connection in the AFT-decomposition remark, and the sensitivity-analysis framing in the SCM section.
- **“To be verified”** flag inside the conditional-unconfoundedness weakening of randomisation remark (Section 1).

- **Two Supplementary-Materials sections on the structural model** (`sm:scm` and `sm:scm-full`). The reduced and expanded versions still overlap.
- **Merged-draft posterior-computation Supplementary-Materials section** (`sm:posterior-computation`). Streamlined replacement for D and F kept alongside the originals; decide which to keep before circulation.
- **Robust-MAP citation placeholder** ([CITE:Schmidli et al. 2014]) in the Supplementary Materials for the heterogeneity hyperprior. Schmidli 2014 is not yet in the bib.
- **General-to-do block** at the foot of the file is the author’s working notes. Remove before circulation.

Logical next steps.

1. **Write out Section 6 (Posterior inference).** Expand the four itemised topics into prose: treatment-effect estimands (CATE, AF, possibly TE-VIM), heterogeneity diagnostics, borrowing diagnostics (σ_d , c -shrinkage), and residual-fit diagnostics (active mixture clusters, F_0 vs F_1).
2. **Resolve the SCM Supplementary-Materials overlap.** Either merge `sm:scm` and `sm:scm-full` into one staged section, or sharpen the distinct roles so the reader is not tempted to read both.
3. **Resolve the posterior-computation Supplementary-Materials overlap.** Either keep the merged streamlined version (G) and delete D + F, or keep D + F and delete G.
4. **Citation to add to the bib.** Schmidli et al. 2014 for the robust MAP heterogeneity prior, and replace the corresponding [CITE:...] marker.
5. **Final stylistic sweep on Section 6 and the SCM Supplementary Materials.** Sections 1–5 have been audited for `-ing` openers, em-dashes, prose semicolons, and multi-comma sentences; the remaining material has not.

GENERAL TO DO

- very thorough check of section Prior specification.
 - Think about Posterior inference;
 - check structural causal model section in the Supplementary Materials thoroughly.
 - Check for other references for the paragraph on the between-study-heterogeneity of the CATE, and our relaxation of the confoundedness assumption.
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